

Public Transport Usage Analysis (2019 & 2022)

1. Dataset Description

Two datasets were analyzed—. The 2019 dataset contains daily totals for buses, trams, and metro: passenger counts with peak/off-peak splits, ticket prices, and mode-level revenues per day. The 2022 dataset comprises individual trip records with fields for date, transport mode, distance (km), duration (minutes), and price (€). Both datasets contain date fields enabling day-wise trends, weekly patterns, and price–distance–duration analyses. The first dataset is aggregated by day and mode; the second contains individual journey data which make it appropriate for regression and dispersion analysis.

2. Figures

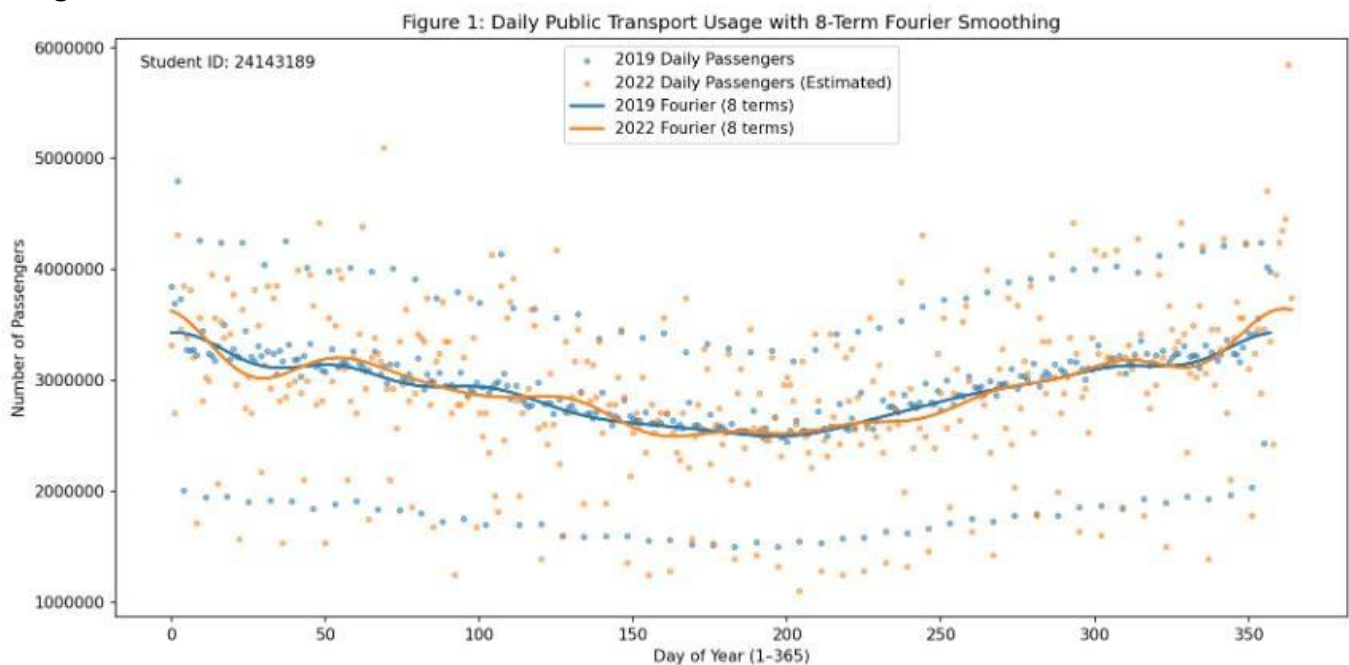


Figure 1. Daily passengers (2019 vs 2022) with 8-term Fourier smoothing. Scatter points show raw day-level usage; the 8-term Fourier curve reveals seasonality, holiday peaks/valleys, and a slight overall rise in 2022 relative to 2019.

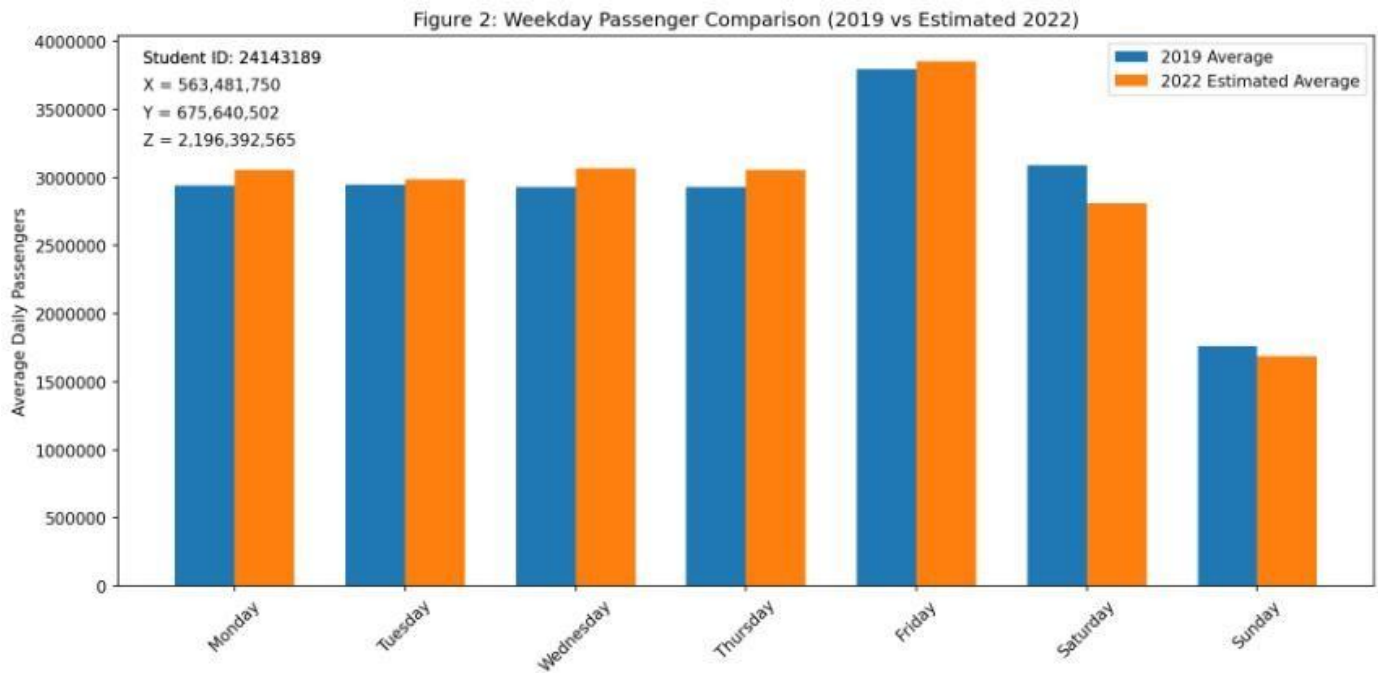


Figure 2. Weekday passenger averages (2019 vs 2022). Bar charts indicate Fridays have the highest ridership in both years; weekday averages are higher in 2022, consistent with travel shifts and possible service improvements.

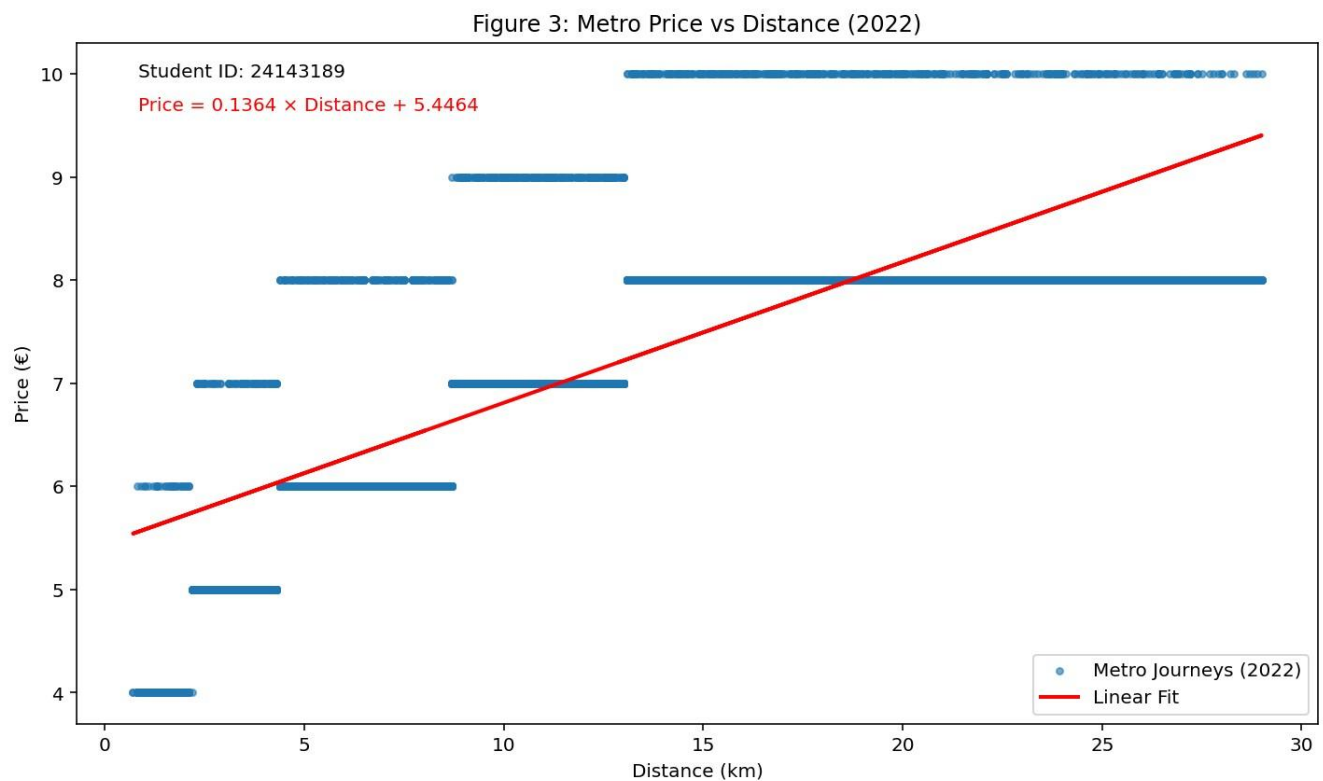


Figure 3. Metro price vs distance (2022) with linear regression. A linear fit shows price increases with distance; estimated base fare $\approx$ €5.45 and distance rate $\approx$ €0.1364 per km.

Figure 4: Journey Distance vs Duration (2022)

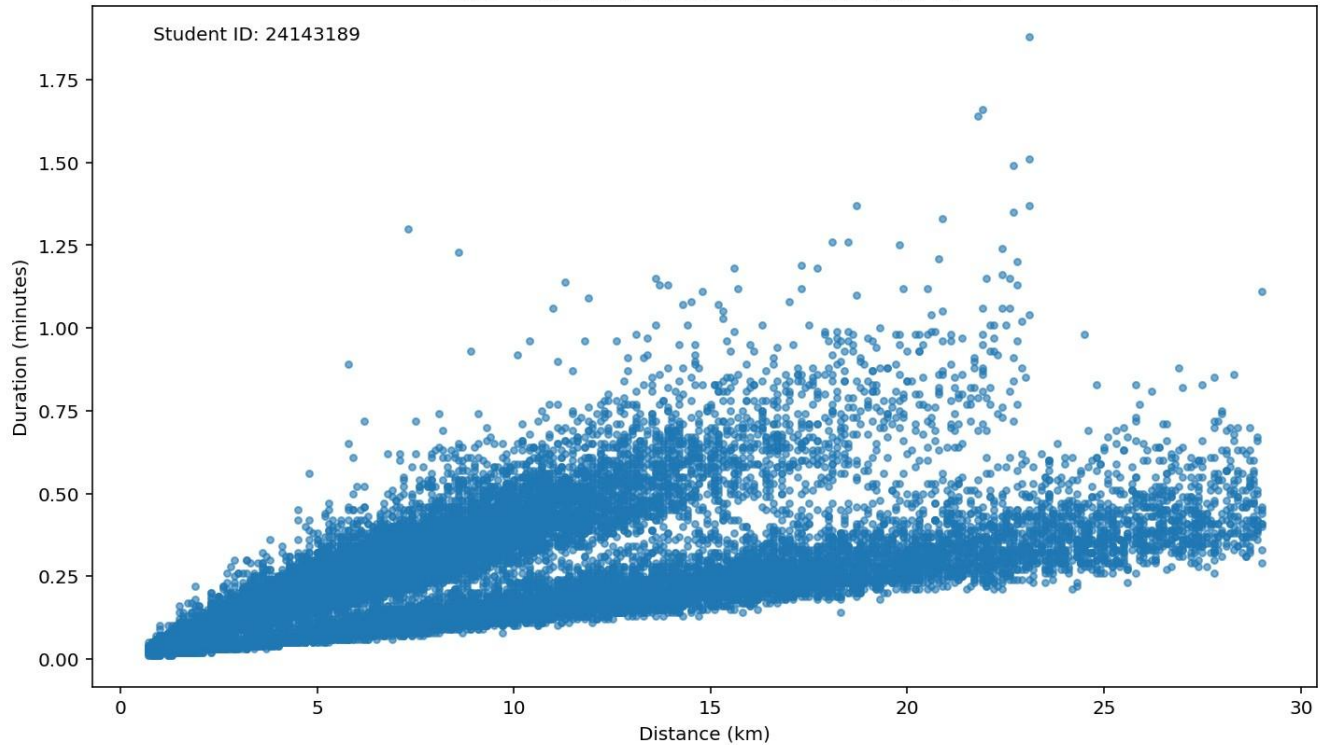


Figure 4. Journey distance vs duration (2022). Durations generally rise with distance, but variability grows for longer trips—likely due to routing, stops, congestion, and delays.

3. Mathematical Formulas

Fourier Series (Smoothed Trend):

$S(t) = a_0 + \sum [a_n * \cos(2\pi n t / N) + b_n * \sin(2\pi n t / N)]$ for $n = 1$ to 8 where:

$S(t)$ = smoothed signal at time t a_0 , a_n ,

b_n = Fourier coefficients from `fft()` t = time

points (day of year)

N = total number of data points

Linear Regression (Price vs Distance):

$$P = \beta_1 * d + \beta_0$$

where:

P = price (€) d =

distance (km)

Revenue Calculations:

$R_{\text{mode}} = (p_{\text{peak}} * c_{\text{peak}}) + (p_{\text{off}} * c_{\text{off}})$ where:

R_{mode} = revenue for transport mode p_{peak} ,

p_{off} = peak and off-peak prices c_{peak} , c_{off} =

peak and off-peak passenger counts

Total Mode Revenues (2019):

$X = \sum R_{\text{bus}} = \text{€}563,481,750$, $Y = \sum R_{\text{trams}} = \text{€}675,640,502$

$Z = \sum R_{\text{metro}} = \text{€}2,196,392,565$

4. Discussion & Conclusions

Daily & weekly variation. Daily passenger counts exhibit pronounced irregularities aligned with holidays and special events; the smoothed Fourier curves highlight recurring seasonal structure across the year. 2022 usage broadly mirrors 2019 but is slightly higher overall, indicating recovery and growth after disruptions. Fridays lead weekly peaks in both years, consistent with commuter and leisure travel patterns; weekday averages are generally higher in 2022.

Monthly & annual patterns. Although figures emphasize day/weekday aggregation, the Fourier trend implies repeating monthly/seasonal components and a year-over-year uplift in 2022 relative to 2019, consistent with service reliability improvements and evolving travel behavior. These periodic components suggest that policy or timetable changes should be evaluated against seasonal baselines rather than single-month snapshots.

Prices & distance. The 2022 metro data shows a stable linear tariff: a material base fare plus a modest per.km increment. This structure explains why longer trips cost more, and it supports predictable budgeting for riders and planners. The fitted coefficients ($\approx \text{€}5.45$ base; $\approx \text{€}0.1364/\text{km}$) quantify this relationship and enable scenario testing, such as fare policy simulations.

Travel time & distance. Durations rise with distance but display wide dispersion at longer ranges, reflecting operational and network factors (stop density, interchange penalties, congestion, and delays). This dispersion implies that distance alone is insufficient for reliable journey_time forecasts; mode_specific and routing variables should be included in performance models.

Revenues & modes. 2019 mode revenues (Bus, Tram, Metro) show metro's dominant share (Z) versus bus (X) and tram (Y), consistent with higher capacity and demand on rapid transit corridors. When combined with 2022's ridership uplift, planners might expect post.2022 revenue increases, especially where metro networks absorb peak demand.

Reasons for differences (2019 vs 2022). The slight rise in 2022 usage is plausibly linked to recovery, service enhancements, and shifts in commuting/leisure patterns. Weekly peaks (Fridays) indicate enduring work.leisure overlap; holiday effects still drive daily spikes/dips. Variability in travel times underscores the need for operational efficiency (reducing delays, optimizing stops and interchanges).

Overall, the figures collectively suggest a system recovering and adapting to new demand profiles, with pricing and capacity aligned to distance-driven journeys.